

LTA analysis - Descriptive Stats

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Reading in dataset from 2009

2009 dataset now includes a total of 8,702 observations (matching analytic dataset in published paper) and includes a total of 79 variables - including indexno

```
lca_pcp_2009 <- read.table(  
  file.path(todd_dir, "lca_pcp_2009_1.24.23.dat"),  
  sep = ",",  
  header = FALSE,  
  fill = TRUE,  
  comment.char = "",  
  quote = ""  
)  
  
#Changing all values of -9999 to missing  
lca_pcp_2009[lca_pcp_2009==-9999] <- NA
```

Reading in dataset from 2019

2019 dataset now includes a total of 7,929 observations (matching analytic dataset in published paper) and includes a total of 79 variables - including indexno

```
lca_pcp_2019 <- read.table(  
  file.path(todd_dir, "lca_pcp2019_1.24.23.dat"),  
  sep = ",",  
  header = FALSE,  
  fill = TRUE,  
  comment.char = "",  
  quote = ""
```

```
)

#Changing all values of -9999 to missing
lca_pcp_2019[lca_pcp_2019==-9999] <- NA

names(lca_pcp_2019) <- vars_2009
```

Reformatting variables for descriptive tables

Now we need to format the variables to be categorical variables in the 2009 dataset

```
lca_pcp_2009 <- lca_pcp_2009 %>%
  mutate(specialty = case_when(
    group_fam    == 1 ~ "Family Medicine",
    group_int    == 1 ~ "General Internal Medicine",
    group_obgyn  == 1 ~ "General Obstetrics and Gynecology",
    group_ped    == 1 ~ "General Pediatrics",
    group_ger    == 1 ~ "Geriatrics",
    TRUE         ~ "Missing"
  ),
  employment_setting = case_when(
    setting_amb  == 1 ~ "Ambulatory Care",
    setting_hosp == 1 ~ "Hospital",
    setting_medsch == 1 ~ "Medical School/Parent University",
    setting_nurs == 1 ~ "Nursing Home/Extended Care",
    setting_other == 1 ~ "Other",
    setting_solo == 1 ~ "Solo practioner",
    setting_missing == 1 ~ "Missing",
    TRUE         ~ "Missing Value"
  ),
  urban_rural = case_when(
    ruca_urban  == 1 ~ "Urban",
    ruca_lrural == 1 ~ "Large Rural",
    ruca_srural == 1 ~ "Small Rural",
    TRUE        ~ "Missing"
  ),
  race_ethnicity = case_when(
    white == 1 ~ "Non-Hispanic White",
    black == 1 ~ "Non-Hispanic Black",
    american_native == 1 ~ "American Indian/Alaskan Native",
    asian_or_pi == 1 ~ "Asian/Pacific Islander",
```

```

    hispanicrace == 1 ~ "Hispanic",
    other_race == 1 ~ "Other",
    race_unknown == 1 ~ "Unknown",
    TRUE ~ "Missing"
  ),
  generational = case_when(
    gen_x == 1 ~ "Gen X (1965-1980)",
    gen_y == 1 ~ "Gen Y (1981-2000)",
    gen_boom_lead == 1 ~ "Baby Boomer, Leading Edge (1946-1954)",
    gen_boom_trail == 1 ~ "Baby Boomer, Trailing Edge (1955-1964)",
    gen_ww2 == 1 ~ "WWII (1928-1945)",
    TRUE ~ "Missing"
  ))

```

Now we need to format the variables to be categorical variables

```

lca_pcp_2019 <- lca_pcp_2019 %>%
  mutate(specialty = case_when(
    group_fam == 1 ~ "Family Medicine",
    group_int == 1 ~ "General Internal Medicine",
    group_obgyn == 1 ~ "General Obstetrics and Gynecology",
    group_ped == 1 ~ "General Pediatrics",
    group_ger == 1 ~ "Geriatrics",
    TRUE ~ "Missing"
  ),
  employment_setting = case_when(
    setting_amb == 1 ~ "Ambulatory Care",
    setting_hosp == 1 ~ "Hospital",
    setting_medsch == 1 ~ "Medical School/Parent University",
    setting_nurs == 1 ~ "Nursing Home/Extended Care",
    setting_other == 1 ~ "Other",
    setting_solo == 1 ~ "Solo practioner",
    setting_missing == 1 ~ "Missing",
    TRUE ~ "Missing Value"
  ),
  urban_rural = case_when(
    ruca_urban == 1 ~ "Urban",
    ruca_lrural == 1 ~ "Large Rural",
    ruca_srural == 1 ~ "Small Rural",
    TRUE ~ "Missing"
  ),
  race_ethnicity = case_when(

```

```

white == 1 ~ "Non-Hispanic White",
black == 1 ~ "Non-Hispanic Black",
american_native == 1 ~ "American Indian/Alaskan Native",
asian_or_pi == 1 ~ "Asian/Pacific Islander",
hispanicrace == 1 ~ "Hispanic",
other_race == 1 ~ "Other",
race_unknown == 1 ~ "Unknown",
TRUE ~ "Missing"
),
generational = case_when(
  gen_x == 1 ~ "Gen X (1965-1980)",
  gen_y == 1 ~ "Gen Y (1981-2000)",
  gen_boom_lead == 1 ~ "Baby Boomer, Leading Edge (1946-1954)",
  gen_boom_trail == 1 ~ "Baby Boomer, Trailing Edge (1955-1964)",
  gen_ww2 == 1 ~ "WWII (1928-1945)",
  TRUE ~ "Missing"
))

```

Creating merged dataset

First we want to rename all variables, except indexno - our ID - with subscript denoting year. Also adding an indicator variable to denote whether the indexno is in 2009 and/or in 2019.

```

lca_pcp_2009_renamed <- lca_pcp_2009 %>%
  rename_with(~ paste0(.x, "_2009"), -indexno, -in_2009) %>%
  mutate(in_2009 = 1L)

lca_pcp_2019_renamed <- lca_pcp_2019 %>%
  rename_with(~ paste0(.x, "_2019"), -indexno, -in_2019) %>%
  mutate(in_2019 = 1L)

```

Now merging dataset - including all indexno in **either** 2009 and/or 2019

```

lca_pcp_merged <- lca_pcp_2009_renamed %>%
  full_join(lca_pcp_2019_renamed, by = "indexno") %>%
  mutate(
    in_2009 = coalesce(in_2009, 0L),
    in_2019 = coalesce(in_2019, 0L)
  )

```

Now looking at descriptive statistics

```
table(lca_pcp_merged$in_2009, lca_pcp_merged$in_2019)
```

```
      0      1  
0      0 3312  
1 4085 4617
```

```
library(dplyr)  
library(tidyr)  
library(stringr)  
  
# numeric variables only (drop IDs + indicators)  
num_vars <- lca_pcp_merged %>%  
  select(where(is.numeric)) %>%  
  select(-indexno, -in_2009, -in_2019) %>%  
  names()  
  
summary_table <- lca_pcp_merged %>%  
  select(indexno, in_2009, in_2019, all_of(num_vars)) %>%  
  pivot_longer(  
    cols = all_of(num_vars),  
    names_to = "variable",  
    values_to = "value"  
  ) %>%  
  mutate(  
    year = case_when(  
      str_ends(variable, "_2009") ~ "2009",  
      str_ends(variable, "_2019") ~ "2019"  
    ),  
    base_var = str_remove(variable, "_2009|_2019")  
  ) %>%  
  # determine binary vs non-binary at base-variable level  
  group_by(base_var) %>%  
  mutate(  
    is_binary = all(na.omit(value) %in% c(0, 1))  
  ) %>%  
  ungroup() %>%  
  # define the four analysis groups  
  mutate(  
    #
```

```

group = case_when(
  in_2009 == 1 & in_2019 == 0 & year == "2009" ~ "2009_only",
  in_2009 == 1 & in_2019 == 1 & year == "2009" ~ "both_2009",
  in_2009 == 1 & in_2019 == 1 & year == "2019" ~ "both_2019",
  in_2009 == 0 & in_2019 == 1 & year == "2019" ~ "2019_only",
  TRUE ~ NA_character_
) %>%
filter(!is.na(group)) %>%
group_by(base_var, is_binary, group) %>%
summarise(
  mean = mean(value, na.rm = TRUE),
  sd    = sd(value, na.rm = TRUE),
  .groups = "drop"
) %>%
# apply formatting rules
mutate(
  value_display = if_else(
    is_binary,
    # binary → percent, rounded to nearest %
    paste0(round(mean * 100), "%"),
    # non-binary → mean (SD), 1 decimal
    paste0(
      round(mean, 1),
      " (",
      round(sd, 1),
      ")"
    )
  )
) %>%
select(base_var, group, value_display) %>%
pivot_wider(
  names_from = group,
  values_from = value_display
) %>%
arrange(base_var)

summary_table

```

```

# A tibble: 78 x 5
  base_var   `2009_only` `2019_only` both_2009 both_2019
  <chr>      <chr>      <chr>      <chr>      <chr>

```

1	_0to5	8%	22%	6%	0%
2	_11to20	26%	27%	39%	21%
3	_21to30	24%	11%	28%	39%
4	_31more	25%	10%	12%	40%
5	_6to10	17%	29%	15%	0%
6	adi_state	4.8 (3)	4.4 (3)	4.6 (3)	4.5 (3)
7	age	47.9 (12.3)	40.8 (10.9)	45.3 (9.2)	55.3 (9.2)
8	age35below	19%	39%	15%	0%
9	age36to45	27%	36%	39%	15%
10	age46to55	24%	13%	32%	39%

i 68 more rows

Formatting this table so this can be output - including column sample sizes

```
n_2009_only <- sum(lca_pcp_merged$in_2009 == 1 & lca_pcp_merged$in_2019 == 0)
n_both <- sum(lca_pcp_merged$in_2009 == 1 & lca_pcp_merged$in_2019 == 1)
n_2019_only <- sum(lca_pcp_merged$in_2009 == 0 & lca_pcp_merged$in_2019 == 1)

summary_table_n <- summary_table

colnames(summary_table_n) <- c(
  "Variable",
  paste0("2009 only (N = ", n_2009_only, ")"),
  paste0("Both years - 2009 (N = ", n_both, ")"),
  paste0("Both years - 2019 (N = ", n_both, ")"),
  paste0("2019 only (N = ", n_2019_only, ")")
)
```

Output summary table

```
write.csv(
  summary_table_n,
  file = "lca_pcp_summary_table_041423.csv",
  row.names = FALSE
)
```